AnalystLab Africa

Machine Learning Internship Programme
Week 2: Business Understanding Report
Data Preprocessing and Feature Engineering for Machine Learning

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1 Executive Summary

This report restates the business context established in Week 1 and translates it into the concrete data preprocessing objectives pursued in Week 2. ABC Communications Ltd, a telecommunications provider, is experiencing customer churn that erodes recurring revenue. The Week 1 phase framed this as a supervised binary classification problem: predict whether a customer will churn, using the Telco Customer Churn dataset. Week 2 carries this objective forward into the preprocessing stage, where the raw dataset is inspected, cleaned, enriched with engineered features, encoded, scaled, and reduced to a machine-learning-ready form, with every decision justified against the business objective rather than performed mechanically.

2 Business Context

2.1 Recap of the Business Problem

Customer churn directly reduces ABC Communications' monthly recurring revenue and increases customer acquisition costs, since retaining an existing subscriber is materially cheaper than acquiring a new one. The business goal is to identify customers at high risk of churning in time for retention teams to intervene with targeted offers or service improvements.

2.2 Why Preprocessing Matters for This Goal

A predictive model is only as reliable as the data it is trained on. The raw dataset used in this project contains a data type inconsistency, hidden missing values, redundant category labels, and a mix of nominal, ordinal, and continuous features that must each be handled appropriately before any model can be trained. Skipping or mishandling this stage would silently degrade the resulting churn predictions and, by extension, the business decisions built on top of them.

3 Machine Learning Problem Definition

3.1 Problem Statement

Given a set of customer account, service subscription, and billing attributes, predict whether the customer will churn (`Churn = Yes`) or remain (`Churn = No`) within the observed billing period. This remains, as established in Week 1, a binary classification task.

3.2 Target Variable

The target variable is `Churn`, encoded during preprocessing as 1 for `Yes` and 0 for `No`. Of the 7,043 customers in the dataset, 5,174 (73.5%) did not churn and 1,869 (26.5%) did churn. This class imbalance is a relevant consideration for model selection and evaluation in later weeks, though it does not affect the preprocessing steps documented here.

3.3 Candidate Input Features

Excluding the identifier column `customerID` (dropped, as it carries no predictive signal) and the target `Churn`, the dataset provides 19 raw candidate features spanning four groups:

- **Demographic:** `gender`, `SeniorCitizen`, `Partner`, `Dependents`
- **Account:** `tenure`, `Contract`, `PaperlessBilling`, `PaymentMethod`
- **Services:** `PhoneService`, `MultipleLines`, `InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`
- **Billing:** `MonthlyCharges`, `TotalCharges`

Section 4 documents the additional engineered features created from this raw set to better expose lifecycle and bundling patterns relevant to churn.

4 Data Quality Issues Identified

The Week 2 inspection phase surfaced the following concrete data quality issues, each carried into the corresponding preprocessing decision documented in the companion Data Preprocessing Report:

1. **Incorrect data type:** `TotalCharges` is stored as text rather than a numeric type, which silently masks missing values from a standard null check.
2. **Hidden missing values:** Coercing `TotalCharges` to numeric reveals 11 rows (0.16% of the dataset) with missing values. All 11 correspond to customers with `tenure = 0`, i.e. customers who had just signed up.
3. **Redundant categorical labels:** Six service columns encode `"No internet service"` and one column encodes `"No phone service"` as separate categories, even though this information is already fully captured by `InternetService` and `PhoneService` respectively.
4. **No duplicate records:** Both full-row duplicates and duplicate `customerID` values were checked for and none were found, so no rows required removal on this basis.
5. **No numeric outliers:** Interquartile range and Z-score analysis of `tenure`, `MonthlyCharges`, and `TotalCharges` found zero outliers by either method. This is a genuine property of the dataset rather than a gap in the analysis, since all three variables are naturally bounded by realistic business limits.

5 Alignment with Business Objective

Every preprocessing and feature engineering decision in Week 2 was evaluated against a single question: does this step make the churn signal easier for a model to learn, without introducing information a model would not have access to at prediction time? For example, the `TenureGroup`, `NumStreamingServices`, and `NumSecurityServices` features engineered in this phase are motivated directly by the retention levers identified in Week 1 (contract length, service bundling, and account age are all actionable dimensions ABC Communications can use in a retention campaign), and the feature importance analysis in Part 5 of the accompanying notebook confirms that contract type and tenure are indeed the strongest predictors of churn in this dataset, consistent with the original business framing.

6 Conclusion

The business problem and target variable defined in Week 1 remain unchanged. Week 2 confirms, through direct inspection of the data, that the dataset is fundamentally sound (no duplicates, no numeric outliers) but requires targeted cleaning (a data type correction and a small, explainable imputation), category standardization, and feature engineering before it is suitable for model training. These steps are documented in full, with justification for every decision, in the Data Preprocessing Report and the accompanying Jupyter notebook.